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Tentative title: Periodogram Averaging with a Sliding Window: Equivalence of the Blackman-Tukey and Bartlett Methods for Spectral Estimation

August 17, 2001

while there are many ways -- still standard pedagogical approach, & most used --
Refer to N' Model

1 Introduction

Classical spectral estimation theory is concerned with estimating the power spectra of stationary random signals in the absence of parametrized signal generator models. Standard treatments divide the subject into variations on two basic methods. The first, commonly known as spectral smoothing or the Blackman-Tukey method, estimates the power spectrum by convolving the raw spectrum of a single finite length signal segment with a smoothing kernel; the second, known as periodogram averaging or Bartlett's method, estimates by sampling over the spectra of multiple smaller sample signals. Conventional engineering wisdom maintains that these methods essentially "do the same thing." However, to our knowledge there have been no previous published reports demonstrating an explicit equivalence. In this communication, we demonstrate by straightforward calculation that, under certain conditions, by smoothing raw spectra and periodogram averaging are formally identical.

2 Problem statement

We state the classical spectral estimation problem as follows. Consider a stationary random signal with time and frequency domain representations $x(t) \leftrightarrow x(\omega)$, respectively, and power spectral density

$$P(\omega) = \langle x(\omega)x^*(\omega) \rangle_x, \quad (1)$$

where "*" denotes the complex conjugate and $\langle \cdot, \cdot \rangle_x$ averaging over an ensemble of signal realizations. Given a segment of length T of a *single* signal realization $x_T(t)$, compute an estimate $\hat{P}(\omega)$ of the power spectrum.

use a different symbol
No - $E[x(\omega)x^*(\omega')] = P(\omega)\delta(\omega-\omega')$

3 Two standard methods

3.1 Smoothing (Blackman-Tukey method)

In the Blackman-Tukey method, one first directly transforms the raw signal segment to the frequency domain, then multiplies by the complex conjugate to obtain the *periodogram* $|x_T(\omega)|^2$. The power spectrum is then computed by convolving the periodogram with an appropriately normalized smoothing kernel $W(\omega)$, producing the estimate

$$\hat{P}_{BT}(\omega) = W(\omega) * |x_T(\omega)|^2. \quad (2)$$

Choosing the precise shape and width of the smoothing kernel involves tradeoffs between various figures of merit for the estimate such as resolution, bias and variance. Smoothing replaces coefficients in the high-resolution raw spectrum with local frequency averages. For this to be appropriate, the signal must of course be bandlimited and the data (signal segment) must represent an over-sampling of the signal.

3.2 Periodogram averaging (Bartlett's method)

The second common approach is that of averaging short periodograms. In this method, one first parses the raw data signal $x_T(t)$ into N non-overlapping segments $\{x_n(t), n = 1, \dots, N\}$. The length of the segments is chosen again according to criteria which trade off such figures as resolution, bias and variance. For example, fewer but longer segments yields less biased estimates with higher variance. The short segments are individually Fourier transformed and multiplied by their complex conjugates, yielding N short periodograms $\{|x_n(\omega)|^2, n = 1, \dots, N\}$. The spectral estimate is obtained by averaging over the short periodograms

$$\hat{S}_{PA}(\omega) = \frac{1}{N} \sum_{n=1}^N |x_n(\omega)|^2. \quad (3)$$

3.3 Modern refinements

Welch introduced the most commonly used refinement to the method of averaging periodograms in a classic paper titled "The Use of the Fast Fourier Transform for the Estimation of Power Spectra: A Method Based on Time Averaging Over Short, Modified Periodograms." Welch's two main contributions to the topic were: (1) the suggestion to modify the signal segments by windowing before computing ~~computing~~ the Fourier transform; this reduces spectral leakage. (2) allowing the signal segments to overlap (usually by 50%); this reduces the estimate variance.

The Blackman-Tukey method may be fine tuned by carefully choosing smoothing kernels to optimize various statistical properties of the estimate mentioned above (Papoulis ref). We will not discuss the choice of smoothing kernel here.

Instead, below we show that by averaging periodograms with arbitrarily large overlap, we are actually applying a smoothing kernel in the frequency domain. This method, which is seen to be a natural extension of Welch's method, provides a bridge between the standard smoothing and periodogram averaging methods of nonparametric spectral estimation.

4 Periodogram averaging with a sliding window

We now show how the Blackman-Tukey method may be obtained as a limiting case of the method of periodogram averaging. We begin by formalizing the discussion from section 3.1.

We desire to estimate the function $P(\omega) = \langle x(\omega)x^*(\omega) \rangle_x$, an average over an *ensemble* of signal realizations $x(t) \leftrightarrow w(\omega)$. However, by assumption, we have access to only *one* realization $x_T(t) \leftrightarrow x_T(\omega)$. Therefore, we approximate $P(\omega)$ by averaging over samples derived from $x_T(t)$. To this end, we introduce a *sampling window* $w(t)$, which we take to be a symmetric function $w(t) = w(-t)$ that is well localized in time. The simplest window is the ideal lowpass or "box" filter; this is effectively the window used to generate segments in the usual periodogram averaging method. However, the box filter is poorly localized in the frequency domain (it "rings"), leading to spectral leakage. This problem is addressed by Welch's suggestion to use *tapered* windows. For our purposes, we will assume that $w(t)$ is gaussian.

N samples are generated by applying the gaussian window to the segment $x_T(t)$ N times, shifted by amounts $\{t'_n = n\Delta; n = 0, \dots, N-1\}$, where $\Delta = T/N$. We denote the samples by $\{s(t, t'_n)\}$. Explicitly,

$$s(t, t'_n) = x_T(t)w(t - t'_n) \leftrightarrow s(\omega, t'_n) = \frac{1}{2\pi} x_T(\omega) * (w(\omega)e^{-j\omega t'_n}). \quad (4)$$

From these samples, we rewrite the average in (3) as

$$\hat{S}(\omega) = \langle s(\omega, t'_n)s^*(\omega, t'_n) \rangle_{t'_n} = \frac{1}{N} \sum_{n=0}^{N-1} |s(\omega, t'_n)|^2, \quad (5)$$

where $\langle \cdot, \cdot \rangle_{t'_n}$ indicates averaging over the sample signals centered at positions $\{t'_n\}$.

The number of samples or, equivalently, the magnitude of the shifting parameter Δ sets the degree of overlap between adjacent samples. Increasing the number of samples is expected to decrease the variance of the estimate. Thus, averaging overlapping segments makes more complete use of the data. However, the variance does not decrease as $1/\sqrt{N}$ because the overlapping segments are correlated. Indeed, we find in numerical experiments with Gaussian random signals an exponentially decreasing improvement as the percent overlap ranges from 0 to 100% (see figure 1). The choice of 50% overlap used in most applications of Welch's method represents a compromise between the decrease in

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variance and increased computational time in practical applications that come with increasing numbers of signal segments.

Following above reasoning, the estimate variance will be minimized by allowing the segment overlap to increase without bound, i.e. by letting $N \rightarrow \infty$, $\Delta = T/N \rightarrow 0$. In this limit, t'_n becomes a continuous variable $t'_n \rightarrow t'$, so that the average in turn becomes

$$\begin{aligned}
 \hat{P}(\omega) &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |s(\omega, t'_n)|^2 \\
 &= \lim_{N \rightarrow \infty} \frac{1}{N\Delta} \sum_{n=0}^{N-1} |s(\omega, t'_n)|^2 \Delta \\
 &= \frac{1}{T} \int_0^T |s(\omega, t')|^2 dt' \\
 &\doteq \langle s(\omega, t') s^*(\omega, t') \rangle_{t'}
 \end{aligned} \tag{6}$$

Substituting explicitly for $s(\omega, t')$, we obtain the following chain:

$$\begin{aligned}
 \langle s(\omega, t') s^*(\omega, t') \rangle_{t'} &= \int dt' s(\omega, t') s^*(\omega, t') \\
 &= \frac{1}{(2\pi)^2} \int dt' \left[\frac{1}{2\pi} x_T(\omega) * (e^{-i\omega t'} w(\omega)) \right] \cdot \left[\frac{1}{2\pi} x_T^*(\omega) * (e^{+i\omega t'} w^*(\omega)) \right] \\
 &= \frac{1}{(2\pi)^2} \int \int \int dt' d\omega' d\omega'' x_T(\omega') e^{-i(\omega - \omega')t'} w(\omega - \omega') x_T^*(\omega'') e^{i(\omega - \omega'')t'} w^*(\omega - \omega'') \\
 &= \frac{1}{(2\pi)^2} \int \int d\omega' d\omega'' x_T(\omega') x_T^*(\omega'') w(\omega - \omega') w^*(\omega - \omega'') \int dt' e^{-i(\omega' - \omega'')t'} \\
 &= \frac{1}{(2\pi)^2} \int \int d\omega' d\omega'' x_T(\omega') x_T^*(\omega'') w(\omega - \omega') w^*(\omega - \omega'') \delta(\omega' - \omega'') \\
 &= \frac{1}{(2\pi)^2} \int d\omega' x_T(\omega') x_T^*(\omega') |w(\omega - \omega')|^2 \\
 &= \frac{1}{(2\pi)^2} |w(\omega)|^2 * [r^*(\omega) s(\omega)].
 \end{aligned} \tag{7}$$

Or, redefining the smoothing kernel $W(\omega) \doteq \frac{1}{(2\pi)^2} |w(\omega)|^2$, we have the simple expression

$$\hat{P}(\omega) = W(\omega) * |x_T(\omega)|^2, \tag{8}$$

which is formally identical to the expression for the Blackman-Tukey estimate, (2). Thus, periodogram averaging with samples collected using a *sliding window* is equivalent to applying a smoothing kernel to the raw spectral data, where the smoothing kernel is proportional to the squared Fourier transform of the sampling window.

5 Conclusions

We have investigated some connections between two standard, classical methods of nonparametric spectral estimation, namely the Blackman-Tukey and periodogram averaging methods. We have shown by a simple, instructive calculation that in the limit as the overlap between adjacent signal segments is allowed grow arbitrarily large, these two methods are formally equivalent. This “sliding window” method is seen to be a natural extension of the now standard method introduced by Welch.

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$$w(t) = \begin{cases} 1, & t \in [-t', t'] \\ 0, & \text{otherwise} \end{cases}.$$

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Or, redefining the smoothing kernel $W(\omega) \doteq \frac{1}{(2\pi)^2} |w(\omega)|^2$, we have the simple expression

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Notes on Spectral Estimation

Brandon Westover

January 12, 2001

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1 A noisy spectral estimator: the periodogram

Given a finite length signal segment $x_T(t)$ extending over time $[-T, T]$ of a single realization of a random process $x(t)$, we wish to estimate the power spectral density $S(\omega)$ of $x(t)$. The power spectrum (PSD) is the Fourier transform (FT) of the autocorrelation function $R(\tau)$:

$$\begin{aligned} S(\omega) &= \mathfrak{F}[R(t)] \\ R(t) &= \mathfrak{F}^{-1}[S(\omega)]. \end{aligned}$$

Accordingly, we can approach spectral estimation in two complementary ways:

1. *Time domain approach*: Estimate $R(\tau)$, then compute its FT.

2. *Frequency domain approach:* Estimate $S(\omega)$ directly.

An obvious starting point using the first approach is to form the following time-average estimate of the autocorrelation function:

$$R^T(\tau) = \frac{1}{2T - |\tau|} \int_{-T+|\tau|/2}^{T-|\tau|/2} x(t + \tau/2)x(t - \tau/2)dt; \quad (1)$$

which for convenience later in this discussion can be rewritten as

$$\begin{aligned} R^T(\tau) &= \frac{1}{2T - |\tau|} \int_{-T+|\tau|}^T x(t)x(t - \tau)dt \\ &= \frac{1}{2T - |\tau|} \int_{-\infty}^{\infty} x_T(t)x_T(-[\tau - t])dt \\ &= \frac{1}{2T - |\tau|} x_T(\tau) * x(-\tau). \end{aligned} \quad (2)$$

(Of course, we let $R^T(\tau) = 0$ for $|\tau| > 2T$.) This estimate is unbiased, as

$$\begin{aligned} E[R^T(\tau)] &= \frac{1}{2T - |\tau|} \int_{-T+|\tau|/2}^{T-|\tau|/2} E[x(t + \tau/2)x(t - \tau/2)]dt \\ &= R(\tau) \frac{1}{2T - |\tau|} \int_{-T+|\tau|/2}^{T-|\tau|/2} dt \\ &= R(\tau), \end{aligned} \quad (3)$$

where $R(\tau)$ is the true autocorrelation function. However, the variance of the estimator is unacceptable as $|\tau| \rightarrow 2T$, due to the decreasing amount of overlap between $x(t + \tau/2)$ and $x(t - \tau/2)$. We can reduce the variance by applying the triangular window $(1 - \frac{|\tau|}{2T})$ to our estimate of the autocorrelation, obtaining the modified estimate $R_T(\tau)$:

$$R_T(\tau) = (1 - \frac{|\tau|}{2T})R^T(\tau) \quad (4)$$

$$= (1 - \frac{|\tau|}{2T})(\frac{1}{2T - |\tau|})x_T(\tau) * x(-\tau) \quad (5)$$

$$= \frac{1}{2T}x_T(\tau) * x(-\tau). \quad (6)$$

The windowed estimate is obviously biased (i.e., $E[R_T(\tau)] = (1 - \frac{|\tau|}{2T})R(\tau) \neq R(\tau)$), but it has smaller variance near the “edges” since these are deemphasized by the window. The particular choice of window is motivated by considering the complimentary first-pass PSD estimate in the frequency domain, the *periodogram*. The periodogram is simply the power of the Fourier transform of

$x_T(t)$, divided by $2T$ (to make the estimate a *density*):

$$\begin{aligned} S_T(\omega) &= \frac{1}{2T} \int_{-\infty}^{\infty} x_T(t) e^{-j\omega t} dt \\ &= \frac{1}{2T} |X_T(\omega)|^2. \end{aligned} \quad (7)$$

The reason for choosing the particular form of the window (triangular) becomes apparent when we compute the FT of $R_T(\tau)$:

$$\begin{aligned} \Im[R_T(t)] &= \Im\left[\frac{1}{2T} x_T(\tau) * x(-\tau)\right] \\ &= \frac{1}{2T} X_T(\omega) X_T(-\omega) \\ &= \frac{1}{2T} X_T(\omega) X_T^*(\omega) \\ &= \frac{1}{2T} |X_T(\omega)|^2. \\ &= S_T(\omega). \end{aligned} \quad (8)$$

That is, the triangular window ensures that our estimated autocorrelation and power spectrum are FT's of one another, just as the true functions are.

Some elementary performance measures of estimators are the bias and variance.

1.1 Bias of the periodogram

The periodogram is a biased estimator, due to the finite length of the signal segment $x_T(t)$:

$$\begin{aligned} E[S_T(\omega)] &= E\left[\int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) R^T(\tau) e^{-j\omega t} dt\right] \\ &= \int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) R(\tau) e^{-j\omega t} dt \\ &= 2T \Im[\rho(\tau) R(\tau)] \\ &= \frac{2T \left[\frac{\sin(T\omega)}{T\omega}\right]^2 * S(\omega)}{2\pi} \\ &= \frac{\sin^2(T\omega)}{\pi T \omega^2} * S(\omega); \end{aligned} \quad (10)$$

that is, mean of the periodogram is a smoothed version of the true PSD. If the data segment is very long, i.e. if $T \rightarrow \infty$, then the smoothing kernel approximates a delta function. So, for large T , we have

$$\begin{aligned} E[S_T(\omega)] &\approx S(\omega) * \delta(\omega) \\ &= S(\omega), \end{aligned} \quad (11)$$

which means that periodogram is *asymptotically* unbiased.

1.2 Variance of the periodogram

Even for long data records $x_T(t)$, the periodogram is an unreliable estimator: its standard deviation at most frequencies (except near $\omega = 0$) is as large as its mean, which is essentially the function we wish to estimate; i.e.,

$$\text{Var}[S_T(\omega)] \approx S^2(\omega). \quad (12)$$

This is of course due to the fact that $S_T(\omega)$ is based on a single sample signal, $x_T(t)$. Note also that collecting a *longer* sample segment will not decrease the variance. In fact, the longer $x_T(t)$ is, the noisier the estimate looks, since increasing T increases the resolution of the periodogram, moving independent frequencies closer together (see Kay p.69). In order to reduce the variance, one must average many periodograms or smooth a single periodogram by convolving it with a smoothing kernel. Smoothing and averaging are closely related operations. (One objective of these notes is to explore and clarify this relationship.)

1.2.1 Variance derivation part I: *groundwork*

The derivation which follows is long and tedious. I include it because it brings together several useful results for the analysis of random signals. To see the derivation of the variance, skip directly to section 1.2.2.

(i) For a stationary random signal $x(t)$ with autocorrelation function (ACF) $R(t_1, t_2) = R(t_1 - t_2)$, the 2-D FT $\Gamma(u, v)$ is

$$\begin{aligned} \Gamma(u, v) &= \int \int R(t_1 - t_2) e^{-j(ut_1 - vt_2)} dt_1 dt_2 \\ &= \int e^{-j(u+v)t_2} \int R(\tau) e^{-ju\tau} d\tau dt_2 \\ &= S(u) \int e^{-j(u+v)t_2} dt_2 \\ &= 2\pi S(u) \delta(u + v). \end{aligned} \quad (13)$$

(ii) If $y(t) = x(t)w(t)$, then the ACF $R_{yy}(\tau)$ of $y(t)$ can be written in terms of the ACF $R_{xx}(\tau)$ of $x(t)$ as

$$\begin{aligned} R_{yy}(t_1, t_2) &= E[y(t_1)y^*(t_2)] \\ &= E[x(t_1)w(t_1)x^*(t_2)w^*(t_2)] \\ &= R_{xx}(t_1 - t_2)w(t_1)w^*(t_2). \end{aligned} \quad (14)$$

(iii) Using (ii), we find:

$$\begin{aligned} \Gamma_{yy}(u, v) &= \mathfrak{F}[R_{xx}(t_1 - t_2)w(t_1)w^*(t_2)] \\ &= \left(\frac{1}{2\pi}\right)^2 \mathfrak{F}[w(t_1)w^*(t_2)] * \mathfrak{F}[R(t_1 - t_2)] \\ &= \left(\frac{1}{2\pi}\right)^2 [W(u)W^*(-v)] * [2\pi S(u)\delta(u + v)] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2\pi} \int \int W(u - \alpha) W^*(-v + \beta) S(\alpha) \delta(\alpha + \beta) d\alpha d\beta \\
&= \frac{1}{2\pi} \int W(u - \beta) W^*(-v - \beta) S(\beta) d\beta.
\end{aligned} \tag{15}$$

(iv) The ACF of the FT of $X(\omega)$ of a random signal $x(t)$ given in terms of its 2-D FT is

$$\begin{aligned}
E[X(u)X^*(v)] &= E\left[\left(\int x(t_1)e^{-jut_1} dt_1\right)\left(\int x(t_2)e^{-jvt_2} dt_2\right)^*\right] \\
&= \int \int E[x(t_1)x^*(t_2)] e^{-jut_1} e^{+jvt_2} dt_1 dt_2 \\
&= \int \int R(t_1 - t_2) e^{-jut_1} e^{+jvt_2} dt_1 dt_2 \\
&= \Gamma(u, -v).
\end{aligned} \tag{16}$$

(v) Putting results (iii) and (iv) together, we see that

$$\begin{aligned}
E[Y(u)Y^*(v)] &= \Gamma_{yy}(u, -v) \\
&= \frac{1}{2\pi} \int W(u - \beta) W^*(v - \beta) S(\beta) d\beta.
\end{aligned} \tag{17}$$

(vi) From (v),

$$\begin{aligned}
E[|Y(\omega)|^2] &= \frac{1}{2\pi} \int |W(\omega - \beta)|^2 S(\beta) d\beta \\
&= |W(\omega)|^2 * S(\omega).
\end{aligned} \tag{18}$$

(vii) The last and most useful result is a formula for the covariance of $|X(\omega)|^2$. Strictly speaking, the formula only apply's for zero-mean, normal random signals. I'll give the formula first, then the (rather long) derivation

$$Cov\{|X(u)|^2, |X(v)|^2\} = \Gamma^2(u, -v) + \Gamma^2(u + v). \tag{19}$$

Derivation of (19):

Before deriving (19), we first fix some notation:

$$\begin{aligned}
X(\omega) &= A(\omega) + jB(\omega) \\
\Gamma(u, v) &= \Gamma_r(u, v) + j\Gamma(u, v).
\end{aligned}$$

We also note from (iv) that if $x(t)$ is real, then $X(-\omega) = X^*(\omega)$, and

$$E[X(u)X(v)] = \Gamma(u, v). \tag{20}$$

Now, using (20) and (16) from (iv) we obtain,

$$\begin{aligned}
\Gamma(u, v) &= \Gamma_r(u, v) + j\Gamma(u, v) \\
&= E\{[A(u) + jB(u)][A(v) + jB(v)]\} \\
&= (E[A(u)A(v)] - E[B(u)B(v)]) + j(E[A(u)B(v)] + E[A(v)B(u)]) \\
\Gamma(u, -v) &= \Gamma_r(u, -v) + j\Gamma(u, -v) \\
&= E\{[A(u) + jB(u)][A(v) - jB(v)]\} \\
&= (E[A(u)A(v)] + E[B(u)B(v)]) + j(-E[A(u)B(v)] + E[A(v)B(u)])
\end{aligned}$$

From (21) and (22) we obtain the four relationships

$$2E[A(u)A(v)] = \Gamma_r(u, v) + \Gamma_r(u, -v) \quad (23)$$

$$2E[A(v)B(u)] = \Gamma_i(u, v) + \Gamma_i(u, -v) \quad (24)$$

$$2E[B(u)B(v)] = -\Gamma_r(u, v) + \Gamma_i(u, -v) \quad (25)$$

$$2E[A(u)B(v)] = \Gamma_i(u, v) - \Gamma_i(u, -v). \quad (26)$$

Finally, we recall (pull out of a magic hat) the result that for jointly normal, zero mean random variables x and y ,

$$E[x^2y^2] = E[x^2]E[y^2] + 2E^2[xy]. \quad (27)$$

Adopting the abbreviated notation $A(u) = A$, $B(u) = B$; $A(v) = A'$, $B(v) = B'$; and $\Gamma(u, v) = \Gamma_r + \Gamma_i$, $\Gamma(u, -v) = \Gamma'_r + \Gamma'_i$, we can use results (23)-(27) to obtain

$$\begin{aligned} \text{Cov}\{|X(u)|^2, |X(v)|^2\} &= E[(|X(u)|^2 - E[|X(u)|^2])(|X(v)|^2 - E[|X(v)|^2])] \\ &= E[|X(u)|^2|X(v)|^2] - E[|X(u)|^2]E[|X(v)|^2] \\ &= E[(A^2 + B^2)(A'^2 + B'^2)] - E[A^2 + B^2]E[A'^2 + B'^2] \\ &= E[A^2A'^2] + E[A^2B'^2] + E[A'^2B^2] + E[B^2B'^2] \\ &\quad - (E[A^2]E[A'^2] + E[A^2]E[B'^2] + E[A'^2]E[B^2] + E[B^2]E[B'^2]) \\ &= 2E^2[AA'] + 2E^2[BB'] + 2E^2[AB'] + 2E^2[A'B] \\ &= \frac{1}{2}\{(2E[AA'])^2 + (2E[BB'])^2 + (2E[AB'])^2 + (2E[A'B])^2\} \\ &= \frac{1}{2}\{(\Gamma_r + \Gamma'_r)^2 + (-\Gamma_r + \Gamma'_r)^2 + (\Gamma_i + \Gamma'_i)^2 + (\Gamma_i - \Gamma'_i)^2\} \\ &= \frac{1}{2}\{2(\Gamma_r^2 + \Gamma_i^2) + 2(\Gamma_r'^2 + \Gamma_i'^2)\} \\ &= \Gamma^2(u, v) + \Gamma^2(u, -v), \end{aligned}$$

as promised.

1.2.2 Variance derivation, part II: *derivation proper*

We can derive equation (12) for the variance of the periodogram using the results of section 1.2.1. We let $x(t) = p_T(t)x(t)$, where $p_T(t)$ and its FT are

$$\begin{aligned} p_T(t) &= \begin{cases} 1 & -T < t < T \\ 0 & \text{otherwise} \end{cases} \\ P_T(\omega) &= \frac{2 \sin(\omega T)}{\omega}. \end{aligned}$$

Now, letting $w(t) = p_T(t)$ in equation (15) and using the fact that we are dealing with real signals, we have

$$\Gamma(u, v) = \frac{1}{2\pi} \int W(u - \beta)W^*(-v - \beta)S(\beta)d\beta$$

$$\begin{aligned}
&= \frac{1}{2\pi} \int W(u - \beta)W(v + \beta)S(\beta)d\beta \\
&= \frac{1}{2\pi} \int W(\beta)W^*(v + [u - \beta])S(u - \beta)d\beta \\
&= \frac{1}{2\pi} \int \frac{2 \sin(\beta T)}{\beta} \frac{2 \sin([u + v - \beta]T)}{[u + v - \beta]} S(u - \beta)d\beta \\
&= \int \frac{2 \sin(\beta T) \sin([u + v - \beta]T)}{\pi \beta [u + v - \beta]} S(u - \beta)d\beta. \tag{28}
\end{aligned}$$

To find the variance, we recall

$$Var[S_T(\omega)] = Cov\{|S_T(\omega)|^2, |S_T(\omega)|^2\}, \tag{29}$$

and then use (28) to compute the two terms required in (19):

$$\begin{aligned}
\Gamma(\omega, -\omega) &= \int \frac{2 \sin(\beta T) \sin(-\beta T)}{\pi \beta [-\beta]} S(\omega - \beta)d\beta \\
&\approx S(\omega) \int \frac{2 \sin^2(\beta T)}{\pi \beta^2} d\beta \\
&= \frac{2\pi T S(\omega)}{\pi} \\
&= 2TS(\omega), \tag{30}
\end{aligned}$$

where the second line is justified by considering that, for large T and $\omega \gg 1/T$, $\frac{2 \sin^2(\beta T)}{\pi \beta^2} \rightarrow \delta(\beta)$; and

$$\begin{aligned}
\Gamma(\omega, \omega) &= \int \frac{2 \sin(\beta T) \sin([2\omega - \beta]T)}{\pi \beta [2\omega - \beta]} S(\omega - \beta)d\beta \\
&\approx 0, \tag{31}
\end{aligned}$$

since for large T , $\omega \gg 1/T$ the fraction in the integrand of (31) behaves essentially like two out-of-phase delta functions. Plugging (30), (31), and (7) into (29), we finally obtain equation (12):

$$\begin{aligned}
Var[S_T(\omega)] &= Cov\{|S_T(\omega)|^2, |S_T(\omega)|^2\} \\
&= \frac{1}{4T^2} Cov\{|X_T(\omega)|^2, |X_T(\omega)|^2\} \\
&= \frac{1}{4T^2} [\Gamma^2(\omega, -\omega) + \Gamma^2(\omega, \omega)] \\
&\approx \frac{1}{4T^2} [(2TS(\omega))^2 + 0] \\
&= S^2(\omega).
\end{aligned}$$

2 Smoothed periodogram

To reduce the variance of the periodogram, we introduce a smoothing kernel or window, $w(t) \leftrightarrow W(\omega)$, and compute the smoothed periodogram

$$\begin{aligned} S_w(\omega) &= \frac{1}{2\pi} S_T(\omega) * W(\omega) \\ &= \Im[R_T(\tau)w(\tau)]. \end{aligned}$$

Some terminology:

- lag window

$$w(t) = \frac{1}{2\pi} \int W(\omega) e^{j\omega t} d\omega$$

- spectral window

$$W(\omega) = \int w(t) e^{-j\omega t} dt$$

We assume that the window has the following properties:

- $w(0) = \int w(t) e^{-j\omega t} d\omega = 1$,
- $W(\omega) \geq 0$,
- $W(-\omega) = W(\omega)$.

In general, the bias and variance of the estimated PSD will depend on the particular form of the window, as well as the length of the data segment $x_T(t)$.

2.1 Bias of the smoothed periodogram

The smoothed periodogram is a biased estimator, even in limit of large T :

$$\begin{aligned} E[S_w(\omega)] &= \frac{1}{2\pi} \int E[S_T(\alpha)] W(\omega - \alpha) d\alpha \\ &= \frac{1}{2\pi} \frac{\sin^2(T\omega)}{\pi T \omega^2} * S(\omega) * W(\omega) \\ &\approx \frac{1}{2\pi} \delta(\omega) * S(\omega) * W(\omega) \\ &= \frac{1}{2\pi} S(\omega) * W(\omega). \end{aligned} \tag{32}$$

2.2 Variance of the smoothed periodogram

To facilitate derivation of a general approximate expression for the variance the smoothed periodogram, we assume:

1. T is large enough that $S(\omega)$ and $W(\omega)$ are nearly constant in any interval of length $df = 1/T$.

2. The width σ of the spectral window $W(\omega)$ is small enough that $S(\omega)$ is essentially constant in any interval of length 2σ .

Also, we generalize equation (28):

$$\begin{aligned}
\Gamma(u, -v) &= \int \frac{2 \sin(\beta T) \sin([u - v - \beta]T)}{\pi \beta [u - v - \beta]} S(u - \beta) d\beta \\
&\approx S(u) \int \frac{2 \sin(\beta T) \sin([u - v - \beta]T)}{\pi \beta [u - v - \beta]} d\beta \\
&= S(u) 2\pi \left[\frac{\sin(T[u - v])}{\pi [u - v]} * \frac{\sin(T[u - v])}{\pi [u - v]} \right] \\
&= 2S(u) \frac{\sin(T[u - v])}{[u - v]}, \tag{33}
\end{aligned}$$

where we have used the fact that the fraction only takes significant values when the arguments of the $\sin(\cdot)$ functions are close to 1; in particular, when $\beta T \approx 1$ and consequently, for large T , $\beta \approx 1/T \approx 0 \Rightarrow S(u - \beta) \approx S(u)$. Similarly,

$$\begin{aligned}
\Gamma(u, v) &= \int \frac{2 \sin(\beta T) \sin([u + v - \beta]T)}{\pi \beta [u + v - \beta]} S(u - \beta) d\beta \\
&\approx S(u) \int \frac{2 \sin(\beta T) \sin([u + v - \beta]T)}{\pi \beta [u + v - \beta]} d\beta \\
&\approx 0. \tag{34}
\end{aligned}$$

Plugging (33) and (34) into (19) yields

$$\begin{aligned}
Cov\{S_T^2(u), S_T^2(v)\} &= \frac{1}{4T^2} Cov\{|X_T(\omega)|^2, |X_T(\omega)|^2\} \\
&= \frac{1}{4T^2} [\Gamma^2(\omega, -\omega) + \Gamma^2(\omega, \omega)] \\
&\approx \frac{1}{4T^2} \left[\left(2S(u) \frac{\sin(T[u - v])}{[u - v]} \right)^2 + 0 \right] \\
&= \frac{\sin^2(T[u - v])}{T^2 [u - v]^2} S^2(u) \tag{35}
\end{aligned}$$

For large T , the fraction in (35) behaves like a delta function, except for a normalization constant. That is,

$$Cov\{S_T^2(u), S_T^2(v)\} = \delta(u - v) q(u).$$

The function $q(u)$ is determined by the normalization property of the delta function:

$$\begin{aligned}
q(u) &= \int \delta(u - v) q(u) dv \\
&= \int Cov\{S_T^2(u), S_T^2(v)\} dv
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{\sin^2(T[u-v])}{T^2[u-v]^2} S^2(u) dv \\
&= \int \frac{\sin^2(Tv)}{(Tv)^2} S^2(u-v) dv \\
&\approx S^2(u) \int \frac{\sin^2(Tv)}{(Tv)^2} dv \\
&= S^2(u) \frac{\pi}{T}.
\end{aligned} \tag{36}$$

Finally, we derive ~~can write compute~~ the (approximate) variance:

$$\begin{aligned}
Var[S_w(\omega)] &= E \{ (S_w(\omega) - E[S_w(\omega)])^2 \} \\
&= E \left\{ \left(\frac{W(\omega - \alpha)}{2\pi} * [S_T(\omega) - E[S_T(\omega)]] \right)^2 \right\} \\
&= \int \int \frac{W(\omega - \alpha)}{2\pi} \frac{W(\omega - \beta)}{2\pi} Cov[S_T(\alpha), S_T(\beta)] d\alpha d\beta \\
&\approx \int \int \frac{W(\omega - \alpha)}{2\pi} \frac{W(\omega - \beta)}{2\pi} S^2(\alpha) \frac{\pi}{T} \delta(\alpha - \beta) d\alpha d\beta \\
&= \frac{1}{2T} \frac{1}{2\pi} \int W^2(\alpha) S^2(\omega - \alpha) d\alpha \\
&\approx \frac{1}{2T} \frac{1}{2\pi} S^2(\omega) \int W^2(\alpha) d\alpha \\
&= \frac{E_w}{2T} S^2(\omega),
\end{aligned}$$

where

$$E_w = \frac{1}{2\pi} \int W^2(\alpha) d\alpha. \tag{37}$$

2.3 Window selection guidelines

We would like a spectral window which simultaneously minimizes the bias and variance of the spectral estimate $S_w(\omega)$. The bias and variance are

$$Var[S_w(\omega)] = \frac{E_w}{2T} S^2(\omega) \tag{38}$$

$$Bias(\omega) = S(\omega) - \frac{1}{2\pi} S(\omega) * W(\omega). \tag{39}$$

According to (38), to minimize the bias, E_w/T must be small or, equivalently, the window must be thin and concentrated in the time domain $\leftrightarrow$ broad and spread out in the frequency domain. So, without loss of generality, we can assume that

1. $w(t) \rightarrow 0$ as $t \rightarrow 2T$, and

2. $w(t) = 0$ for $|t| > M$, where $M < 2T$.

Recalling the relation between the spectral and lag windows, we now have

$$\begin{aligned} S_w(\omega) &= \int_{-\infty}^{\infty} w(\tau) R_T(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-M}^M w(\tau) R_T(\tau) e^{-j\omega\tau} d\tau \\ &= \mathfrak{F}[w(\tau) R_T(\tau)]. \end{aligned}$$

We that to minimize the bias, the window must be broad and spread out in the time domain $\leftrightarrow$ thin and δ function-like in the frequency domain. An “uncertainty principle” is at play here: the window cannot be simultaneously concentrated in the time and frequency domains. Thus, choosing a window always requires a compromise between bias and variance.

Here are some rules of thumb useful in selecting/designing smoothing windows:

1. $W(\omega) \geq 0$, to ensure positivity.
2. $\int W(\omega) d\omega = 2\pi \leftrightarrow \int w(t) dt = 1$, to ensure consistency. (??)
3. The duration of $W(\omega)$ should be small to ensure a small bias. The second moment m_2 is one common measure of duration:

$$m_2 = \frac{1}{2\pi} \int \omega^2 W(\omega) d\omega.$$

4. The energy E_w of $W(\omega)$ must be small to ensure small variance. Again, this is at odds with requirement (3).
5. The window should have *small sidelobes*, and the height of the sidelobes should decay rapidly to zero, to minimize the effects of spectral leakage. The requirement can be shown equivalent to requiring that as many derivatives as possible should be zeros at the endpoints $\pm M$ of the lag window $w(t)$.

Numerous windows exist in the literature!

3 General windowing issues

3.1 Frequency resolution using the DFT

Two factors limit the frequency resolution of the estimated PSD, that is, how close two peaks can be in the spectrum without smearing together into a single entity. These are:

1. *Number of points in the DFT.* The spectrum produced by an N point DFT consists of evenly spaced samples ranging from minus $1/2$ the sampling frequency to $1/2$ the sampling frequency. To separate two nearby peaks, the sample spacing must be greater than the distance separating the peaks. Adding zeros to the ends of a signal to increase the spectral resolution before computing the DFT is known as “zero padding.”
2. *Length of the input signal.* Any smooth signal viewed for a short enough time will appear to be composed of only a single sinusoid. The closer in frequency two embedded sinusoids are, the longer the signal must be to distinguish them. So the length of the signal T limits the frequency resolution. No amount of zero padding can compensate for insufficient data.

3.2 Spectral Leakage

Computing the DFT of a finite-length signal amounts to projecting the observed signal onto a basis set (complex exponentials) spanning the observation interval. In the preceding discussion, we have implicitly taken for granted that the sinusoids present in the data signal match the frequencies of the basis functions. In such cases, each frequency present in the data projects onto a single sinusoid in the basis set, contributing a single point to the PSD. However, when the input signal contains a sinusoid whose frequency falls *between* two basis function frequencies, this sinusoid makes a nonzero projection onto *all* of the basis functions, i.e. the sinusoid’s true spectrum “leaks” over the entire basis set.

To explore spectral leakage in more detail, we perform the following thought experiment: Imagine that we wish to analyze an infinitely long sinusoid $x(t) = \sin(\omega_0 t)$, using a computer. We do this in two parts.

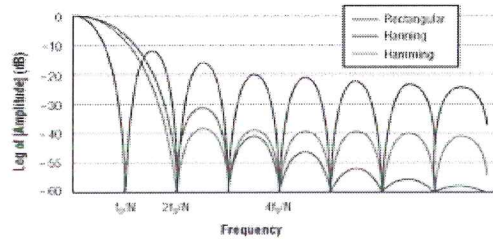
1. *Apply a window to the data.* Obviously, we can only deal with a finite amount of the infinite data. So we multiply signal $x(t)$ by a window $w(t)$ which is nonzero over a finite interval to obtain $x_T(t) = x(t)w(t)$. Now the data is essentially of only finite extent: outside the window, $x_T(t)$ is zero. Next, we compute the FT of $x_T(t)$, $X_T(\omega)$.

$$\begin{aligned} X_T(\omega) &= \mathfrak{F}[x_T(t)] \\ &= \mathfrak{F}[x(t)w(t)] \\ &= \frac{1}{2\pi} X(\omega) * W(\omega). \end{aligned}$$

The sinusoid’s spectrum is a pair of delta functions at $\pm\omega_0$. As a result of windowing the data in the time domain, these impulses are convolved with the window’s Fourier transform $W(\omega)$, placing a copy of $W(\omega)$ at $\pm\omega_0$.

2. *Sample the spectrum.* $X_T(\omega)$ is a continuous signal, consisting in principle of an uncountably infinite number of samples. To enter it into a computer, we must sample the spectrum; i.e., we sample the spectral window.

The window choice determines the nature of the spectral leakage. Choosing a window involves a compromise between bias and resolution (I think the underlying issue essentially the same one we encountered in discussing the tradeoff between bias and variance): A thin mainlobe for $W(\omega)$ provides higher resolution. A rectangular window gives the narrowest mainlobe, thus the best resolution. However, the second lobe of the rectangular window is still large, and the height of subsequent lobes decays slowly, so that spectral leakage is significant. Other window choices (e.g., Blackman-Harris, Welch, Keiser, Chebyshev, Gaussian, Hamming, and Hanning windows, etc) which reduce spectral leakage do so by having smaller second sidelobes, and subsequent sidelobes (generally) which decay more rapidly than for the square window. Three examples are shown in the figure below for comparison.



4 Optimal decoding with a sliding window

For now, I'm only including the math.

4.1 Derivation of the decoding filter

Notation:

signal segment	$x(t) \leftrightarrow X(\omega)$
raw spike train	$r(t) \leftrightarrow R(\omega)$
filter	$\phi(t) \leftrightarrow \Phi(\omega)$
signal estimate	$\hat{x}(t) = \phi(t) * r(t) \leftrightarrow \Phi(\omega)R(\omega)$
window	$w(t) \leftrightarrow W(\omega)$

Define the error ε as:

$$\begin{aligned} \varepsilon &= \left\langle \int dt [x(t) - \hat{x}(t)]^2 \right\rangle \\ &= \left\langle \int dt [x(t) - \phi(t) * r(t)]^2 \right\rangle \end{aligned}$$

$$= \left\langle \int \frac{d\omega}{2\pi} [X(\omega) - \Phi(\omega)R(\omega)]^2 \right\rangle. \quad (40)$$

Now, take the functional derivative wrt $\Phi(\omega)$ and setting it equal to zero, we get

$$\frac{\delta \varepsilon}{\delta \Phi(\omega)} = \left\langle \frac{1}{2\pi} [X(\omega) - \Phi(\omega)R(\omega)]^* [-R(\omega)] \right\rangle = 0,$$

yielding

$$\Phi(\omega) = \frac{\langle X(\omega)R^*(\omega) \rangle}{\langle R(\omega)R^*(\omega) \rangle}. \quad (41)$$

4.2 Explicit formula for the decoding filter

To obtain an explicit formula, we need to specify how to do the averaging. We define the samples as:

- signal samples:

$$\begin{cases} x(t, t') = x(t)w(t - t') \\ X(\omega, t') = \frac{1}{2\pi} X(\omega) * [e^{-j\omega t'} W(\omega)] \end{cases}$$

- spike train samples

$$\begin{cases} r(t, t') = r(t)w(t - t') \\ R(\omega, t') = \frac{1}{2\pi} R(\omega) * [e^{-j\omega t'} W(\omega)] \end{cases}$$

Now we compute the numerator of (41):

$$\begin{aligned} \langle X(\omega)R^*(\omega) \rangle &= \frac{1}{T} \int dt' S(\omega, t') R^*(\omega, t') \quad (2\pi)^2 \\ &= \frac{1}{T} \int dt' \left\{ \frac{1}{2\pi} X(\omega) * [e^{-j\omega t'} W(\omega)] \right\} \left\{ \frac{1}{2\pi} R^*(\omega) * [e^{+j\omega t'} W^*(\omega)] \right\} \\ &= \frac{1}{(2\pi)^2 T} \int dt' \left\{ \int d\omega' X(\omega') e^{-j(\omega - \omega')t'} W(\omega - \omega') \right\} \left\{ \int d\omega'' R^*(\omega'') e^{+j(\omega - \omega'')t'} W^*(\omega - \omega'') \right\} \\ &= \frac{1}{(2\pi)^2 T} \int \int d\omega' d\omega'' X(\omega') R^*(\omega'') W(\omega - \omega') W^*(\omega - \omega'') \int dt' e^{-j(\omega'' - \omega')t'} \\ &= \frac{1}{(2\pi)^2 T} \int \int d\omega' d\omega'' X(\omega') R^*(\omega'') W(\omega - \omega') W^*(\omega - \omega'') [2\pi \delta(\omega' - \omega'')] \\ &= \frac{1}{2\pi T} \int d\omega' X(\omega') R^*(\omega'') W(\omega - \omega') W^*(\omega - \omega') \\ &= \frac{1}{2\pi T} |W(\omega)|^2 * [X(\omega)R(\omega)]. \end{aligned}$$

Doing the same calculation for the denominator, we finally get:

$$\begin{aligned}
\Phi(\omega) &= \frac{\langle X(\omega)R^*(\omega) \rangle}{\langle R(\omega)R^*(\omega) \rangle} \\
&= \frac{|W(\omega)|^2 * [X(\omega)R^*(\omega)]}{|W(\omega)|^2 * |R^*(\omega)|^2}.
\end{aligned} \tag{42}$$

Notes on Nonparametric Spectral Estimation

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1 A noisy spectral estimator: the periodogram

Given a finite length signal segment $x_T(t)$ extending over time $[-T, T]$ of a single realization of a random process $x(t)$, we wish to estimate the power spectral density $S(\omega)$ of $x(t)$. The power spectrum (PSD) is the Fourier transform (FT) of the autocorrelation function $R(\tau)$:

$$\begin{aligned} S(\omega) &= \mathfrak{F}[R(t)] \\ R(t) &= \mathfrak{F}^{-1}[S(\omega)]. \end{aligned}$$

Accordingly, we can approach spectral estimation in two complementary ways:

1. *Time domain approach*: Estimate $R(\tau)$, then compute its FT.

2. *Frequency domain approach:* Estimate $S(\omega)$ directly.

An obvious starting point using the first approach is to form the following time-average estimate of the autocorrelation function:

$$R^T(\tau) = \frac{1}{2T - |\tau|} \int_{-T+|\tau|/2}^{T-|\tau|/2} x(t + \tau/2)x(t - \tau/2)dt; \quad (1)$$

which for convenience later in this discussion can be rewritten as

$$\begin{aligned} R^T(\tau) &= \frac{1}{2T - |\tau|} \int_{-T+|\tau|}^T x(t)x(t - \tau)dt \\ &= \frac{1}{2T - |\tau|} \int_{-\infty}^{\infty} x_T(t)x_T(-[\tau - t])dt \\ &= \frac{1}{2T - |\tau|} x_T(\tau) * x(-\tau). \end{aligned} \quad (2)$$

(Of course, we let $R^T(\tau) = 0$ for $|\tau| > 2T$.) This estimate is unbiased, as

$$\begin{aligned} E[R^T(\tau)] &= \frac{1}{2T - |\tau|} \int_{-T+|\tau|/2}^{T-|\tau|/2} E[x(t + \tau/2)x(t - \tau/2)]dt \\ &= R(\tau) \frac{1}{2T - |\tau|} \int_{-T+|\tau|/2}^{T-|\tau|/2} dt \\ &= R(\tau), \end{aligned} \quad (3)$$

where $R(\tau)$ is the true autocorrelation function. However, the variance of the estimator is unacceptable as $|\tau| \rightarrow 2T$, due to the decreasing amount of overlap between $x(t + \tau/2)$ and $x(t - \tau/2)$. We can reduce the variance by applying the triangular window $(1 - \frac{|\tau|}{2T})$ to our estimate of the autocorrelation obtaining the modified estimate $R_T(\tau)$:

$$R_T(\tau) = (1 - \frac{|\tau|}{2T})R^T(\tau) \quad (4)$$

$$= (1 - \frac{|\tau|}{2T})(\frac{1}{2T - |\tau|})x_T(\tau) * x(-\tau) \quad (5)$$

$$= \frac{1}{2T}x_T(\tau) * x(-\tau). \quad (6)$$

The windowed estimate is obviously biased (i.e., $E[R_T(\tau)] = (1 - \frac{|\tau|}{2T})R(\tau) \neq R(\tau)$), but it has smaller variance near the “edges” since these are deemphasized by the window. The particular choice of window is motivated by considering the complimentary first-pass PSD estimate in the frequency domain, the *periodogram*. The periodogram is simply the power of the Fourier transform of

$x_T(t)$, divided by $2T$ (to make the estimate a *density*):

$$\begin{aligned} S_T(\omega) &= \frac{1}{2T} \int_{-\infty}^{\infty} x_T(t) e^{-j\omega t} dt \\ &= \frac{1}{2T} |X_T(\omega)|^2. \end{aligned} \quad (7)$$

The reason for choosing the particular form of the window (triangular) becomes apparent when we compute the FT of $R_T(\tau)$:

$$\begin{aligned} \Im[R_T(t)] &= \Im\left[\frac{1}{2T} x_T(\tau) * x(-\tau)\right] \\ &= \frac{1}{2T} X_T(\omega) X_T(-\omega) \\ &= \frac{1}{2T} X_T(\omega) X_T^*(\omega) \\ &= \frac{1}{2T} |X_T(\omega)|^2. \\ &= S_T(\omega). \end{aligned} \quad (8)$$

That is, the triangular window ensures that our estimated autocorrelation and power spectrum are FT's of one another, just as the true functions are.

Some elementary performance measures of spectral estimators are the bias and variance.

1.1 Bias of the periodogram

The periodogram is a biased estimator, due to the finite length of the signal segment $x_T(t)$:

$$\begin{aligned} E[S_T(\omega)] &= E\left[\int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) R^T(\tau) e^{-j\omega t} dt\right] \\ &= \int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) R(\tau) e^{-j\omega t} dt \\ &= 2T \Im[\rho(\tau) R(\tau)] \\ &= \frac{2T \left[\frac{\sin(T\omega)}{T\omega}\right]^2 * S(\omega)}{2\pi} \\ &= \frac{\sin^2(T\omega)}{\pi T \omega^2} * S(\omega), \end{aligned} \quad (10)$$

where $\rho(\tau) = \frac{1}{2T} \left(1 - \frac{|\tau|}{2T}\right)$. That is, the mean of the periodogram is a smoothed version of the true PSD. If the data segment is very long, i.e. if $T \rightarrow \infty$, then the smoothing kernel approximates a delta function. So, for large T , we have

$$\begin{aligned} E[S_T(\omega)] &\approx S(\omega) * \delta(\omega) \\ &= S(\omega), \end{aligned} \quad (11)$$

which means that periodogram is asymptotically unbiased.

1.2 Variance of the periodogram

Even for long data records $x_T(t)$, the periodogram is an unreliable estimator: its standard deviation at most frequencies (except near $\omega = 0$) is as large as its mean, which is essentially the function we wish to estimate; i.e.,

$$\text{Var}[S_T(\omega)] \approx S^2(\omega). \quad (12)$$

This is of course due to the fact that $S_T(\omega)$ is based only on a single sample signal, $x_T(t)$. Note also that collecting a *longer* sample segment will not decrease the variance. In fact, the longer $x_T(t)$ is, the noisier the estimate looks, since increasing T increases the resolution of the periodogram, moving independent frequencies closer together (see Kay p.69). In order to reduce the variance, one must average many periodograms or smooth a single periodogram by convolving it with a smoothing kernel. Smoothing and averaging are closely related operations. (One objective of these notes is to explore and clarify this relationship.)

1.2.1 Variance derivation part I: *groundwork*

The derivation which follows is long and tedious. I include it because it brings together several useful results for the analysis of random signals. To see the derivation of the variance, skip directly to section 1.2.2.

(i) For a stationary random signal $x(t)$ with autocorrelation function (ACF) $R(t_1, t_2) = R(t_1 - t_2)$, the 2-D FT $\Gamma(u, v)$ is

$$\begin{aligned} \Gamma(u, v) &= \int \int R(t_1 - t_2) e^{-j(ut_1 - vt_2)} dt_1 dt_2 \\ &= \int e^{-j(u+v)t_2} \int R(\tau) e^{-j u \tau} d\tau dt_2 \\ &= S(u) \int e^{-j(u+v)t_2} dt_2 \\ &= 2\pi S(u) \delta(u + v). \end{aligned} \quad (13)$$

(ii) If $y(t) = x(t)w(t)$, then the ACF $R_{yy}(\tau)$ of $y(t)$ can be written in terms of the ACF $R_{xx}(\tau)$ of $x(t)$ as

$$\begin{aligned} R_{yy}(t_1, t_2) &= E[y(t_1)y^*(t_2)] \\ &= E[x(t_1)w(t_1)x^*(t_2)w^*(t_2)] \\ &= R_{xx}(t_1 - t_2)w(t_1)w^*(t_2). \end{aligned} \quad (14)$$

(iii) Using (ii), we find:

$$\begin{aligned} \Gamma_{yy}(u, v) &= \mathfrak{F}[R_{xx}(t_1 - t_2)w(t_1)w^*(t_2)] \\ &= \left(\frac{1}{2\pi}\right)^2 \mathfrak{F}[w(t_1)w^*(t_2)] * \mathfrak{F}[R(t_1 - t_2)] \\ &= \left(\frac{1}{2\pi}\right)^2 [W(u)W^*(-v)] * [2\pi S(u)\delta(u + v)] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2\pi} \int \int W(u - \alpha) W^*(-v + \beta) S(\alpha) \delta(\alpha + \beta) d\alpha d\beta \\
&= \frac{1}{2\pi} \int W(u - \beta) W^*(-v - \beta) S(\beta) d\beta.
\end{aligned} \tag{15}$$

(iv) The ACF of the FT of $X(\omega)$ of a random signal $x(t)$ given in terms of its 2-D FT is

$$\begin{aligned}
E[X(u)X^*(v)] &= E[(\int x(t_1)e^{-jut_1} dt_1)(\int x(t_2)e^{-jvt_2} dt_2)^*] \\
&= \int \int E[x(t_1)x^*(t_2)] e^{-jut_1} e^{+jvt_2} dt_1 dt_2 \\
&= \int \int R(t_1 - t_2) e^{-jut_1} e^{+jvt_2} dt_1 dt_2 \\
&= \Gamma(u, -v).
\end{aligned} \tag{16}$$

(v) Putting results (iii) and (iv) together, we see that *for $y(t) = x(t)w(t)$,*

$$\begin{aligned}
E[Y(u)Y^*(v)] &= \Gamma_{yy}(u, -v) \\
&= \frac{1}{2\pi} \int W(u - \beta) W^*(v - \beta) S(\beta) d\beta.
\end{aligned} \tag{17}$$

(vi) From (v),

$$\begin{aligned}
E[|Y(\omega)|^2] &= \frac{1}{2\pi} \int |W(\omega - \beta)|^2 S(\beta) d\beta \\
&= \frac{1}{2\pi} |W(\omega)|^2 * S(\omega).
\end{aligned} \tag{18}$$

(vii) The last and most useful result is a formula for the covariance of $|X(\omega)|^2$. Strictly speaking, the formula only applies for zero-mean, normal random signals. I'll give the formula first, then the (rather long) derivation

$$Cov\{|X(u)|^2, |X(v)|^2\} = \Gamma^2(u, -v) + \Gamma^2(u + v). \tag{19}$$

Derivation of (19):

Before deriving (19), we first fix some notation:

$$\begin{aligned}
X(\omega) &= A(\omega) + jB(\omega) \\
\Gamma(u, v) &= \Gamma_r(u, v) + j\Gamma_i(u, v).
\end{aligned}$$

For the sake of readability, we shorten use the abbreviated notation:

$$\left\{ \begin{array}{l} A(u) = A \\ B(u) = B \\ A(v) = A' \\ B(v) = B' \end{array} \right. \text{ and } \left\{ \begin{array}{l} \Gamma(u, v) = \Gamma_r + \Gamma_i \\ \Gamma(u, -v) = \Gamma_r' + \Gamma_i' \end{array} \right.$$

We also note from (iv) that if $x(t)$ is real, then $X(-\omega) = X^*(\omega)$, and

$$E[X(u)X(v)] = \Gamma(u, v). \tag{20}$$

Now, using (20) and (16) from (iv) we obtain,

$$\begin{aligned}\Gamma(u, v) &= \Gamma_r + j\Gamma_i \\ &= E[(A + jB)(A' + jB')] \\ &= (E[AA'] - E[BB']) + j(E[AB'] + E[A'B])\end{aligned}\quad (21)$$

$$\begin{aligned}\Gamma(u, -v) &= \Gamma'_r + j\Gamma'_i \\ &= E[(A + jB)(A' - jB')] \\ &= (E[AA'] + E[BB']) + j(-E[AB'] + E[A'B]).\end{aligned}\quad (22)$$

From (21) and (22) we obtain the four relationships

$$2E[AA'] = \Gamma_r + \Gamma'_r \quad (23)$$

$$2E[A'B] = \Gamma_i + \Gamma'_i \quad (24)$$

$$2E[BB'] = -\Gamma_r + \Gamma'_r \quad (25)$$

$$2E[AB'] = \Gamma_i - \Gamma'_i. \quad (26)$$

Next, we recall (pull out of a magic hat) that for jointly normal, zero mean random variables x and y ,

$$E[x^2y^2] = E[x^2]E[y^2] + 2E^2[xy]. \quad (27)$$

Finally, using results (23)-(27) we obtain

$$\begin{aligned}Cov\{|X(u)|^2, |X(v)|^2\} &= E[(|X(u)|^2 - E[|X(u)|^2])(|X(v)|^2 - E[|X(v)|^2])] \\ &= E[|X(u)|^2|X(v)|^2] - E[|X(u)|^2]E[|X(v)|^2] \\ &= E[(A^2 + B^2)(A'^2 + B'^2)] - E[A^2 + B^2]E[A'^2 + B'^2] \\ &= E[A^2A'^2] + E[A^2B'^2] + E[A'^2B^2] + E[B^2B'^2] \\ &\quad - (E[A^2]E[A'^2] + E[A^2]E[B'^2] + E[A'^2]E[B^2] + E[B^2]E[B'^2]) \\ &= 2E^2[AA'] + 2E^2[BB'] + 2E^2[AB'] + 2E^2[A'B] \\ &= \frac{1}{2}\{(2E[AA'])^2 + (2E[BB'])^2 + (2E[AB'])^2 + (2E[A'B])^2\} \\ &= \frac{1}{2}\{(\Gamma_r + \Gamma'_r)^2 + (-\Gamma_r + \Gamma'_r)^2 + (\Gamma_i + \Gamma'_i)^2 + (\Gamma_i - \Gamma'_i)^2\} \\ &= \frac{1}{2}\{2(\Gamma_r^2 + \Gamma_i^2) + 2(\Gamma_r'^2 + \Gamma_i'^2)\} \\ &= \Gamma^2(u, v) + \Gamma^2(u, -v),\end{aligned}$$

as promised.

1.2.2 Variance derivation, part II

We can derive equation (12) for the variance of the periodogram using the results of section 1.2.1. We let $x(t) = p_T(t)x(t)$, where $p_T(t)$ and its FT are

$$\begin{cases} p_T(t) = \begin{cases} 1 & -T < t < T \\ 0 & \text{otherwise} \end{cases} \\ P_T(\omega) = \frac{2\sin(\omega T)}{\omega} \end{cases}.$$

Now, letting $w(t) = p_T(t)$ in (15) and capitalizing on the fact that we are dealing with *real* signals, we have

$$\begin{aligned}
\Gamma(u, v) &= \frac{1}{2\pi} \int W(u - \beta) W^*(-v - \beta) S(\beta) d\beta \\
&= \frac{1}{2\pi} \int W(u - \beta) W(v + \beta) S(\beta) d\beta \\
&= \frac{1}{2\pi} \int W(\beta) W^*(v + [u - \beta]) S(u - \beta) d\beta \\
&= \frac{1}{2\pi} \int \frac{2 \sin(\beta T)}{\beta} \frac{2 \sin([u + v - \beta]T)}{[u + v - \beta]} S(u - \beta) d\beta \\
&= \int \frac{2 \sin(\beta T) \sin([u + v - \beta]T)}{\pi \beta [u + v - \beta]} S(u - \beta) d\beta. \tag{28}
\end{aligned}$$

To find the variance of $S(\omega)$, we note that

$$Var[S_T(\omega)] = Cov\{|S_T(\omega)|^2, |S_T(\omega)|^2\}, \tag{29}$$

and then use (28) to compute the two terms required in (19):

$$\begin{aligned}
\Gamma(\omega, -\omega) &= \int \frac{2 \sin(\beta T) \sin(-\beta T)}{\pi \beta [-\beta]} S(\omega - \beta) d\beta \\
&\approx S(\omega) \int \frac{2 \sin^2(\beta T)}{\pi \beta^2} d\beta \\
&= \frac{2\pi T S(\omega)}{\pi} \\
&= 2TS(\omega), \tag{30}
\end{aligned}$$

where the second line is justified when T is large and $\omega \gg 1/T$, since then $\frac{2 \sin^2(\beta T)}{\pi \beta^2} \rightarrow \delta(\beta)$. Also,

$$\begin{aligned}
\Gamma(\omega, \omega) &= \int \frac{2 \sin(\beta T) \sin([2\omega - \beta]T)}{\pi \beta [2\omega - \beta]} S(\omega - \beta) d\beta \\
&\approx 0, \tag{31}
\end{aligned}$$

since for large T , $\omega \gg 1/T$ the fraction in the integrand of (31) behaves essentially like two out-of-phase delta functions. Plugging (30), (31), and (7) into (29), we finally obtain equation (12):

$$\begin{aligned}
Var[S_T(\omega)] &= Cov\{|S_T(\omega)|^2, |S_T(\omega)|^2\} \\
&= \frac{1}{4T^2} Cov\{|X_T(\omega)|^2, |X_T(\omega)|^2\} \\
&= \frac{1}{4T^2} [\Gamma^2(\omega, -\omega) + \Gamma^2(\omega, \omega)] \\
&\approx \frac{1}{4T^2} [(2TS(\omega))^2 + 0] \\
&= S^2(\omega).
\end{aligned}$$

2 Smoothed periodogram

To reduce the variance of the periodogram, we introduce a smoothing kernel or window, $w(t) \leftrightarrow W(\omega)$, and compute the smoothed periodogram

$$\begin{aligned} S_w(\omega) &= \frac{1}{2\pi} S_T(\omega) * W(\omega) \\ &= \Im[R_T(\tau)w(\tau)]. \end{aligned}$$

Some terminology from the signal processing literature:

- lag window

$$w(t) = \frac{1}{2\pi} \int W(\omega) e^{j\omega t} d\omega$$

- spectral window

$$W(\omega) = \int w(t) e^{-j\omega t} dt$$

We assume that the window has the following properties:

- $w(0) = \int w(t) e^{-j\omega t} d\omega = 1$,
- $W(\omega) \geq 0$,
- $W(-\omega) = W(\omega)$.

In general, the bias and variance of the estimated PSD will depend on the particular form of the window, as well as the length of the data segment $x_T(t)$.

2.1 Bias of the smoothed periodogram

The smoothed periodogram is a biased estimator, even in limit of large T :

$$\begin{aligned} E[S_w(\omega)] &= \frac{1}{2\pi} \int E[S_T(\alpha)] W(\omega - \alpha) d\alpha \\ &= \frac{1}{2\pi} \frac{\sin^2(T\omega)}{\pi T \omega^2} * S(\omega) * W(\omega) = W_1(\omega) * W_2(\omega) * S(\omega) \\ &\approx \frac{1}{2\pi} \delta(\omega) * S(\omega) * W(\omega) \\ &= \frac{1}{2\pi} S(\omega) * W(\omega). \end{aligned} \tag{32}$$

2.2 Variance of the smoothed periodogram

To facilitate derivation of a general approximate expression for the variance the smoothed periodogram, we assume:

1. T is large enough that $S(\omega)$ and $W(\omega)$ are nearly constant in any interval of length $df = 1/T$.

2. The width σ of the spectral window $W(\omega)$ is small enough that $S(\omega)$ is essentially constant in any interval of length 2σ .

Also, we generalize equation (28):

$$\begin{aligned}
\Gamma(u, -v) &= \int \frac{2 \sin(\beta T) \sin([u - v - \beta]T)}{\pi \beta [u - v - \beta]} S(u - \beta) d\beta \\
&\approx S(u) \int \frac{2 \sin(\beta T) \sin([u - v - \beta]T)}{\pi \beta [u - v - \beta]} d\beta \\
&= S(u) 2\pi \left[\frac{\sin(T[u - v])}{\pi [u - v]} * \frac{\sin(T[u - v])}{\pi [u - v]} \right] \\
&= 2S(u) \frac{\sin(T[u - v])}{[u - v]}, \tag{33}
\end{aligned}$$

where we have used the fact that the fraction only takes significant values when the arguments of the $\sin(\cdot)$ functions are close to 1; in particular, when $\beta T \approx 1$ and consequently, for large T , $\beta \approx 1/T \approx 0 \Rightarrow S(u - \beta) \approx S(u)$. Similarly,

$$\begin{aligned}
\Gamma(u, v) &= \int \frac{2 \sin(\beta T) \sin([u + v - \beta]T)}{\pi \beta [u + v - \beta]} S(u - \beta) d\beta \\
&\approx S(u) \int \frac{2 \sin(\beta T) \sin([u + v - \beta]T)}{\pi \beta [u + v - \beta]} d\beta \\
&\approx 0. \tag{34}
\end{aligned}$$

Plugging (33) and (34) into (19) yields

$$\begin{aligned}
Cov\{S_T^2(u), S_T^2(v)\} &= \frac{1}{4T^2} Cov\{|X_T(\omega)|^2, |X_T(\omega)|^2\} \\
&= \frac{1}{4T^2} [\Gamma^2(\omega, -\omega) + \Gamma^2(\omega, \omega)] \\
&\approx \frac{1}{4T^2} \left[\left(2S(u) \frac{\sin(T[u - v])}{[u - v]} \right)^2 + 0 \right] \\
&= \frac{\sin^2(T[u - v])}{T^2 [u - v]^2} S^2(u) \tag{35}
\end{aligned}$$

For large T , the fraction in (35) behaves like a delta function, except for a normalization constant. That is,

$$Cov\{S_T^2(u), S_T^2(v)\} = \delta(u - v) q(u).$$

The function $q(u)$ is determined by the normalization property of the delta function:

$$\begin{aligned}
q(u) &= \int \delta(u - v) q(u) dv \\
&= \int Cov\{S_T^2(u), S_T^2(v)\} dv
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{\sin^2(T[u-v])}{T^2[u-v]^2} S^2(u) dv \\
&= \int \frac{\sin^2(Tv)}{(Tv)^2} S^2(u-v) dv \\
&\approx S^2(u) \int \frac{\sin^2(Tv)}{(Tv)^2} dv \\
&= S^2(u) \frac{\pi}{T}.
\end{aligned} \tag{36}$$

Finally, we compute the (approximate) variance:

$$\begin{aligned}
Var[S_w(\omega)] &= E \{ (S_w(\omega) - E[S_w(\omega)])^2 \} \\
&= E \left\{ \left(\frac{W(\omega - \alpha)}{2\pi} * [S_T(\omega) - E[S_T(\omega)]] \right)^2 \right\} \\
&= \int \int \frac{W(\omega - \alpha)}{2\pi} \frac{W(\omega - \beta)}{2\pi} Cov[S_T(\alpha), S_T(\beta)] d\alpha d\beta \\
&\approx \int \int \frac{W(\omega - \alpha)}{2\pi} \frac{W(\omega - \beta)}{2\pi} S^2(\alpha) \frac{\pi}{T} \delta(\alpha - \beta) d\alpha d\beta \\
&= \frac{1}{2T} \frac{1}{2\pi} \int W^2(\alpha) S^2(\omega - \alpha) d\alpha \\
&\approx \frac{1}{2T} \frac{1}{2\pi} S^2(\omega) \int W^2(\alpha) d\alpha \\
&= \frac{E_w}{2T} S^2(\omega),
\end{aligned} \tag{37}$$

where

$$E_w = \frac{1}{2\pi} \int W^2(\alpha) d\alpha. \tag{38}$$

2.3 Window selection guidelines

We would like a spectral window which simultaneously minimizes the bias and variance of the spectral estimate $S_w(\omega)$. The bias and variance are

$$\begin{cases} Var[S_w(\omega)] \approx \frac{E_w}{2T} S^2(\omega) \\ Bias(\omega) \approx S(\omega) - \frac{1}{2\pi} S(\omega) * W(\omega) \end{cases} \tag{39}$$

According to (39), to minimize the bias, E_w/T must be small or, equivalently, the window must be thin and concentrated in the time domain $\leftrightarrow$ broad and spread out in the frequency domain. So, without loss of generality, we can assume that

1. $w(t) \rightarrow 0$ as $t \rightarrow 2T$, and
2. $w(t) = 0$ for $|t| > M$, where $M < 2T$.

Recalling the relation between the spectral and lag windows, we now have

$$\begin{aligned} S_w(\omega) &= \mathfrak{F}[w(\tau)R_T(\tau)] \\ &= \int_{-\infty}^{\infty} w(\tau)R_T(\tau)e^{-j\omega\tau}d\tau \\ &= \int_{-M}^M w(\tau)R_T(\tau)e^{-j\omega\tau}d\tau. \end{aligned}$$

We see that to minimize the bias, the window must be broad and spread out in the time domain $\leftrightarrow$ thin and δ -function-like in the frequency domain. An “uncertainty principle” is ~~at~~ operational here: the window cannot be simultaneously concentrated in the time and frequency domains. Thus, choosing a window always requires a compromise between bias and variance.

Here are some rules of thumb useful in selecting/designing smoothing windows:

1. $W(\omega) \geq 0$, to ensure positivity.
2. $\int W(\omega)d\omega = 2\pi \leftrightarrow \int w(t)dt = 1$, to ensure consistency. (??)
3. The duration of $W(\omega)$ should be small to ensure a small bias. The second moment m_2 is one common measure of duration:

$$m_2 = \frac{1}{2\pi} \int \omega^2 W(\omega)d\omega.$$

4. The energy E_w of $W(\omega)$ must be small to ensure small variance. Again, this is at odds with requirement (3).
5. The window should have *small sidelobes*, and the height of the sidelobes should decay rapidly to zero, to minimize the effects of spectral leakage. The requirement can be shown equivalent to requiring that as many derivatives as possible should evaluate to zero at the endpoints $\pm M$ of the lag window $w(t)$.

Numerous windows exist in the literature!

3 General windowing issues

3.1 Frequency resolution using the DFT

Two factors limit the frequency resolution of the estimated PSD, that is, how close two peaks can be in the spectrum without smearing together into a single entity. These are:

1. *Number of points in the DFT.* The spectrum produced by an N point DFT consists of evenly spaced samples ranging from minus $1/2$ the sampling

frequency to $1/2$ the sampling frequency. To separate two nearby peaks, the sample spacing must be greater than the distance separating the peaks. Adding zeros to the ends of a signal to increase the spectral resolution before computing the DFT is known as “zero padding.”

2. *Length of the input signal.* Any smooth signal viewed for a short enough time will appear to be composed of only a single sinusoid. The closer in frequency two embedded sinusoids are, the longer the signal must be to distinguish them. So the length of the signal T limits the frequency resolution. No amount of zero padding can compensate for insufficient data.

3.2 Spectral Leakage

Computing the DFT of a finite-length signals amounts to projecting the observed signal onto a basis set (complex exponentials) spanning the observation interval. In the preceding discussion, we have implicitly taken for granted that the sinusoids present in the data signal match the frequencies of the basis functions. In such cases, each frequency present in the data projects onto a single sinusoid in the basis set, contributing a single point to the PSD. However, when the input signal contains a sinusoid whose frequency falls *between* two basis function frequencies, this sinusoid makes a nonzero projection onto *all* of the basis functions, i.e. the sinusoid’s true spectrum “leaks” over the entire basis set.

To explore spectral leakage in more detail, we perform the following thought experiment: Imagine that we wish to analyze an infinitely long sinusoid $x(t) = \sin(\omega_0 t)$, using a computer. We do this in two parts.

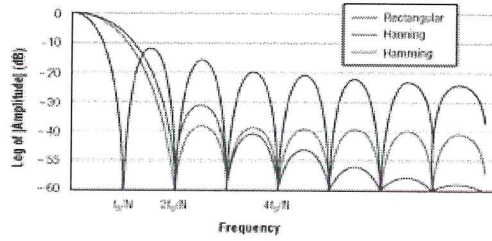
1. *Apply a window to the data.* Obviously, we can only deal with a finite amount of the infinite data. So we multiply signal $x(t)$ by a window $w(t)$ which is nonzero over a finite interval to obtain $x_T(t) = x(t)w(t)$. Now the data is essentially of only finite extent: outside the window, $x_T(t)$ is zero. Next, we compute the FT of $x_T(t)$, $X_T(\omega)$.

$$\begin{aligned} X_T(\omega) &= \mathcal{F}[x_T(t)] \\ &= \mathcal{F}[x(t)w(t)] \\ &= \frac{1}{2\pi} X(\omega) * W(\omega). \end{aligned}$$

The sinusoid’s spectrum is a pair of delta functions at $\pm\omega_0$. As a result of windowing the data in the time domain, these impulses are convolved with the window’s Fourier transform $W(\omega)$, placing a copy of $W(\omega)$ at $\pm\omega_0$.

2. *Sample the spectrum.* $X_T(\omega)$ is a continuous signal, consisting in principle of an uncountably infinite number of samples. To enter it into a computer, we must sample the spectrum; i.e., we sample the spectral window.

The window choice determines the nature of the spectral leakage. Choosing a window involves a compromise between bias and resolution (I think the underlying issue essentially the same one we encountered in discussing the tradeoff between bias and variance): A thin mainlobe for $W(\omega)$ provides higher resolution. A rectangular window gives the narrowest mainlobe, thus the best resolution. However, the second lobe of the rectangular window is still large, and the height of subsequent lobes decays slowly, so that spectral leakage is significant. Other window choices (e.g., Blackman-Harris, Welch, Keiser, Chebyshev, Gaussian, Hamming, and Hanning windows, etc) which reduce spectral leakage do so by having smaller second sidelobes, and subsequent sidelobes (generally) which decay more rapidly than for the square window. Three examples are shown in the figure below for comparison.



4 Optimal decoding with a sliding window

For now, I'm only including the math.

4.1 Derivation of the decoding filter

Notation:

signal segment	$x(t) \leftrightarrow X(\omega)$
raw spike train	$r(t) \leftrightarrow R(\omega)$
filter	$\phi(t) \leftrightarrow \Phi(\omega)$
signal estimate	$\hat{x}(t) = \phi(t) * r(t) \leftrightarrow \Phi(\omega)R(\omega)$
window	$w(t) \leftrightarrow W(\omega)$

Define the error ε as:

$$\begin{aligned}
 \varepsilon &= \left\langle \int dt [x(t) - \hat{x}(t)]^2 \right\rangle \\
 &= \left\langle \int dt [x(t) - \phi(t) * r(t)]^2 \right\rangle \\
 &= \left\langle \int \frac{d\omega}{2\pi} [X(\omega) - \Phi(\omega)R(\omega)]^2 \right\rangle.
 \end{aligned} \tag{40}$$

Now, take the functional derivative wrt $\Phi(\omega)$ and setting it equal to zero, we get

$$\frac{\delta \varepsilon}{\delta \Phi(\omega)} = \left\langle \frac{1}{2\pi} [X(\omega) - \Phi(\omega)R(\omega)]^* [-R(\omega)] \right\rangle = 0,$$

yielding

$$\Phi(\omega) = \frac{\langle X(\omega)R^*(\omega) \rangle}{\langle R(\omega)R^*(\omega) \rangle}. \quad (41)$$

4.2 Explicit formula for the decoding filter

To obtain an explicit formula, we need to specify how to do the averaging. We define the samples as:

- signal samples:

$$\begin{cases} x(t, t') = x(t)w(t - t') \\ X(\omega, t') = \frac{1}{2\pi} X(\omega) * [e^{-j\omega t'} W(\omega)] \end{cases}$$

- spike train samples

$$\begin{cases} r(t, t') = r(t)w(t - t') \\ R(\omega, t') = \frac{1}{2\pi} R(\omega) * [e^{-j\omega t'} W(\omega)] \end{cases}$$

Now we compute the numerator of (41):

$$\begin{aligned} (2\pi)^2 (2T) \langle X(\omega)R^*(\omega) \rangle &= (2\pi)^2 \int dt' S(\omega, t') R(\omega, t') \\ &= \int dt' \{ X(\omega) * [e^{-j\omega t'} W(\omega)] \} \{ R^*(\omega) * [e^{+j\omega t'} W^*(\omega)] \} \\ &= \int dt' \int d\omega' X(\omega') e^{-j(\omega - \omega')t'} W(\omega - \omega') \int d\omega'' R^*(\omega'') e^{+j(\omega - \omega'')t'} W^*(\omega - \omega'') \\ &= \int \int d\omega' d\omega'' X(\omega') R^*(\omega'') W(\omega - \omega') W^*(\omega - \omega'') \int dt' e^{-j(\omega'' - \omega')t'} \\ &= \int \int d\omega' d\omega'' X(\omega') R^*(\omega'') W(\omega - \omega') W^*(\omega - \omega'') [2\pi \delta(\omega' - \omega'')] \\ &= 2\pi \int d\omega' X(\omega') R^*(\omega') W(\omega - \omega') W^*(\omega - \omega') \\ &= 2\pi W(\omega)^2 * [X(\omega)R(\omega)] \end{aligned}$$

Doing the same calculation for the denominator, we finally get:

$$\begin{aligned} \Phi(\omega) &= \frac{\langle X(\omega)R^*(\omega) \rangle}{\langle R(\omega)R^*(\omega) \rangle} \\ &= \frac{|W(\omega)|^2 * [X(\omega)R^*(\omega)]}{|W(\omega)|^2 * |R^*(\omega)|^2}. \end{aligned} \quad (42)$$